Frederik Spors
DS Capstone - Topic
09/07/2026

Research Question: How does watching meaningful versus scrambled human movements affect brain activity across people? *(not final and might be tweaked a bit)*

Advisor: Alison Harris (CMC Neuroscience)

Description: I want to look at how the brain responds when people watch human actions versus random movement (not mimicking human movement). Using EEG data, I will apply inter-subject correlation (ISC) in MATLAB or Python to compare brain activity during coherent versus scrambled movements, and then probably use Bayesian analysis to assess the strength of these differences and their relationship to the mu rhythm.

Advanced Data Analysis: I will analyze EEG data and measure similarities in brain activity across participants. I will use inter-subject correlation (ISC) and probably Bayesian analysis to compare the neural responses to coherent versus scrambled human actions.